Bound of Semi-stable Reduction of Kummer Surfaces

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Editorial note

This document is a mathematical transcription of the six-page handwritten PDF. The original notation has been preserved whenever it is legible. A small number of conventions have been made explicit so that the notes can be read as a self-contained mathematical draft. In particular, bars such as $\overline{P}$ and $\overline{w}$ denote the "adjoint "construction used in the notes, and $W(\Delta)$ denotes the finite abelian group attached to a rational simplex. Statements which were visibly crossed out in the original notes have not been incorporated as assertions.

1 Waterman points and the adjoint construction

The paper draft follows notions and notations referred to in Chapter II, §3 of [**Toroidal Embeddings**]. The quotient M_0/M is finite; its dual description is encoded by the finite abelian group $W(\Delta)$.

Let Δ be a n-simplex with integral vertices $P_0, \ldots, P_n$. Denote by e_i the vector $\overrightarrow{P_0P_i}$ for all $i \geq 1$. For a point P in the affine span of Δ, assuming P_0 the origin, we may slightly abuse notation and write

$$P = \overrightarrow{P_0P} = \sum_{i=1}^{n} a_i e_i. \tag{1}$$

Definition 1.1. A rational structure L on Δ is an abelian group consists of functions with rational values at all P_i. Also L is required to have rank $\dim \Delta$ and contains $q\mathbb{Z}$ for some rational number q. We say L is integral over $\frac{1}{\mu}\mathbb{Z}$ if all functions in L take values in $\frac{1}{\mu}\mathbb{Z}$ at vertices.

Definition 1.2. Assume L is a rational structure on Δ, then the multiplicity of Δ with respect to L and integer μ satisfying that L is integral over $\frac{1}{\mu}\mathbb{Z}$ is defined as

$$m(\Delta, L, \mu) := (\mu)^{\dim \Delta} \cdot (\dim \Delta)! \cdot \mathrm{vol}_L(\Delta) \tag{2}$$

where $\mathrm{vol}_L(\Delta)$ is the integral of the volume form $\varphi_L^*(dx_1 \wedge \cdots \wedge dx_n)$ along Δ, and $\varphi_L : \mathbb{R}^n \to \mathbb{R}^n$ is induced by a $\mathbb{Z}$-basis of L.

Definition 1.3. Denote Γ the lattice spanned by $e_0, \cdots, e_n$ and $W(\Delta) := \mathbb{Z}^{\oplus n}/\Gamma$. Also define

$$M_0 = \{\text{affine functions taking integral values at the vertices}\}, \tag{3}$$

and

$$M = \{\text{coordinate/affine functions taking integral values at all integral points}\}. \tag{4}$$

Then there is a perfect pairing

$$M/M_0 \times W(\Delta) \longrightarrow \mathbb{Q}/\mathbb{Z} \tag{5}$$

and hence we have $m(\Delta, M_0, 1) = \sharp M/M_0 = \sharp W(\Delta)$.

For any $[w] \in W(\Delta)$, it has a unique representative which can written as $\sum_{i=1}^{n} a_i e_i$, where $0 \leq a_i < 1$ for all i. Define the value of $[w]$ to be $v([w]) = \lceil \sum_{i=1}^{n} a_i \rceil$. Obviously, we have $0 \leq v([w]) \leq n$. And there is a map

$$\mathcal{P} : W(\Delta) \setminus \{0\} \longrightarrow \Delta \quad [w] = \sum_{i=1}^{n} a_i e_i \longmapsto \sum_{i=1}^{n} \frac{a_i}{v([w])} e_i \tag{6}$$

We call elements in $\mathrm{im}\,\mathcal{P}$ Waterman points of Δ (with respect to M_0).

Lemma 1.4. *For any Waterman point P of Δ, assume $[w_0] \in \mathcal{P}^{-1}(P)$ such that $v([w_0]) = \min\{v([w])\big|[w] \in \mathcal{P}^{-1}(P)\}$. Then $v([w_0])$ equal to the minimal number v such that $v \cdot P$ is an integral point.*

Definition 1.5. Let P be a Waterman point of value v of Δ. Write $[v \cdot P] = \sum_{i=1}^{n} a_i e_i$ with $0 \leq a_i < 1$. We say P is an interior Waterman point of Δ if it is not contained in any face of Δ, equivalently, $a_i \neq 0$ for all i and $\sum_{i=1}^{n} a_i \neq 1$. Otherwise, we say P is a face Waterman point of Δ.

Definition 1.6 (Adjoint point)**.** Suppose that P is a Waterman point of Δ of value v, and choose a representative

$$P = \sum_i v a_i e_i, \quad [w] = \sum_i v a_i e_i, \quad 0 \leq v a_i < 1. \tag{7}$$

Define

$$[\overline{w}] := \left[\sum_i (1 - v a_i) e_i\right], \quad \overline{P} := \mathcal{P}([\overline{w}]). \tag{8}$$

The point $\overline{P}$ is called the *adjoint point* of P.

Lemma 1.7 (Face/interior compatibility)**.** *The point P is a face Waterman point (resp. an interior Waterman point) if and only if its adjoint point $\overline{P}$ is a face Waterman point (resp. an interior Waterman point).*

Lemma 1.8. *There always exists a Waterman point of value $\leq \lceil \frac{n}{2} \rceil$ in Δ. In particular, if Δ admits any interior Waterman point, then There always exists an interior Waterman point of value $\leq \lceil \frac{n}{2} \rceil$ in Δ.*

2 Rational structures and multiplicity

Now let Δ be a cell in a compact polyhedral complex. Denote by $\widetilde{\Delta}$ a standard simplex isomorphic to Δ, where the isomorphism is determined by a basis of the rational structure on Δ. Write

$$\Delta \xrightarrow[\sim]{(f_1,\ldots,f_n)} \widetilde{\Delta}. \tag{9}$$

Each f_i induces the i-th coordinate projection on $\widetilde{\Delta}$. And in this case, we would define $W(\Delta) := W(\widetilde{\Delta})$ and Waterman points of Δ to be preimages of Waterman points of $\widetilde{\Delta}$.

The notation in the notes is arranged so that a rational structure L on Δ corresponds to the integral coordinate structure M on $\widetilde{\Delta}$. Consequently,

$$m(\Delta, L, \mu) = m(\Delta, \mu L, 1) = m(\widetilde{\Delta}, M, 1). \tag{10}$$

Thus questions about multiplicity may be transferred to the standard simplex. In particular, the multiplicity is

$$m(\Delta, L, \mu) = \#W(\Delta) \tag{11}$$

after the above normalization.

Lemma 2.1 (Prime divisor gives a Waterman point)**.** *Assume that the rational structure L on Δ is integral over $\frac{1}{\mu}\mathbb{Z}$ and that*

$$p \mid m(\Delta, L, \mu) \tag{12}$$

for a prime p. Then Δ contains a Waterman point P of order p.

Proof. Since $W(\Delta)$ is a finite abelian group, the structure theorem gives

$$W(\Delta) \cong \bigoplus_i \mathbb{Z}/p_i^{n_i}\mathbb{Z}. \tag{13}$$

Because

$$p \mid m(\Delta, L, \mu) = m(\widetilde{\Delta}, M, 1) = \#W(\Delta), \tag{14}$$

there is an element $[w] \in W(\widetilde{\Delta})$ of order p. Choose such a class with the value $v([w])$ minimal, and put

$$P := \mathcal{P}([w]). \tag{15}$$

We claim that P has order p.

Suppose instead that

$$\overline{P} = \sum_i a_i e_i \tag{16}$$

has value $v \neq v([w])$. By the definition used in the notes, for every $f \in M_0$ one has an integrality constraint of the form

$$f(P) \in \frac{1}{v([w])\, p}\mathbb{Z}. \tag{17}$$

Equivalently,

$$f(vP) \in \frac{v}{v([w])\, p}\mathbb{Z}. \tag{18}$$

Using the characterization of the value as the smallest positive integer making the corresponding point integral, this forces the order of $[vP]$ to divide p. The minimal choice of $[w]$ rules out the nontrivial alternative, so the order of vP is 1. Hence vP is integral, contradicting the defining property of the Waterman point. Therefore P has order p. □

3 A numerical estimate for a Waterman point

Lemma 3.1. *Let Δ be an n-simplex and let P be a Waterman point of value v and order p in Δ. Then*

$$v < n\,\frac{p-1}{2} + 1. \tag{19}$$

Proof. Write the barycentric coordinates in the form $a_i v$, with

$$a_i v \in \frac{1}{p}\mathbb{Z}, \quad a_i v < 1. \tag{20}$$

Hence

$$a_i v \leq \frac{p-1}{p}. \tag{21}$$

The adjoint construction gives the complementary value

$$\overline{v} = n - v + 1. \tag{22}$$

If necessary replace P by its adjoint so that $v \leq \overline{v}$. Then

$$v - 1 \leq \frac{n(p-1)}{2}, \tag{23}$$

which yields

$$v < n\,\frac{p-1}{2} + 1 \tag{24}$$

in the convention of the notes. □

Remark 3.2 (A caution from the notes). For a single simplex one can introduce a numerical function, with

$$N(p) = 1 + N(p-1), \quad N(1) = 0. \tag{25}$$

However, after subdividing a simplicial complex, compatibility between neighboring simplices becomes essential. This is why the notes emphasize *interior* Waterman points.

4 Compatibility across faces

Lemma 4.1. *There are natural injective group homomorphisms*

$$i_\alpha : W(\Delta_\alpha) \hookrightarrow W(\Delta) \tag{26}$$

where Δ_α is a face of Δ.

For faces $\Delta_\alpha, \Delta_\beta$ of a simplicial complex, the notes warn that in general

$$W(\Delta_\alpha) \cap W(\Delta_\beta) = W(\Delta_{\alpha\beta}), \tag{27}$$

and this group may be nonzero. Only in the particular two-dimensional configuration singled out in the notes does one automatically obtain

$$W(\Delta_{\alpha\beta}) = 0. \tag{28}$$

Thus a construction made independently on each simplex need not give a compatible global subdivision.

5 Prime-by-prime decomposition of the bad locus

Let X be a finite simplicial complex with rational structure L, integral over $\frac{1}{\mu}\mathbb{Z}$, and suppose

$$m(X, L, \mu) = k. \tag{29}$$

For a prime $p \mid k$, define the p-bad locus

$$U_p := \bigcup_{\substack{\sigma \in X \\ p \mid m(\sigma)}} \sigma. \tag{30}$$

Lemma 5.1 (Decomposition of the p-bad locus)**.** *For every prime $p \mid k$, there exist simplices $\sigma_1, \ldots, \sigma_s \in X$ such that*

$$U_p = \bigcup_{i=1}^{s} \overline{\mathrm{St}(\sigma_i)}, \tag{31}$$

no two σ_i, σ_j are contained in a common simplex, and every σ_i admits an interior Waterman point P_i of order p.

Proof. For every simplex Δ with $p \mid m(\Delta)$, Lemma 2.1 supplies a Waterman point of order p. The face maps induce homomorphisms between the corresponding W-groups. If

$$p \nmid m(\Delta_\alpha) = \#W(\Delta_\alpha) \tag{32}$$

for every proper face $\Delta_\alpha \prec \Delta$, then the order-$p$ point cannot lie in the W-group of a proper face. Hence it is an interior Waterman point.

To organize these simplices, set

$$\Sigma_p := \{\sigma \in X : p \mid m(\sigma) \text{ and } p \nmid m(\tau) \text{ for every proper face } \tau \prec \sigma\}. \tag{33}$$

If σ satisfies $p \mid m(\sigma)$ but $\sigma \notin \Sigma_p$, then there is a proper face $\tau < \sigma$ with $p \mid m(\tau)$. Iterating on dimension shows

$$U_p = \bigcup_{\sigma \in \Sigma_p} \overline{\mathrm{St}(\sigma)}. \tag{34}$$

If two elements of Σ_p are contained in a common simplex, discard one of them in the redundant star decomposition. Since X is finite, this process terminates and produces the required family $\sigma_1, \ldots, \sigma_s$. By minimality with respect to faces, each selected σ_i carries an interior Waterman point of order p. □

Remark 5.2. In the last step of discarding simplices, we neglect the case that if there exists σ with $p \mid, (\sigma)$ and there is a unique $\sigma_0 \in \Sigma_p$ such that $\sigma \in \overline{St}(\sigma_0)$. Then if during discarding process, we discard σ_0, then the equality fails. We need to repair this issue.

6 Subdivision step and multiplicity reduction

Proposition 6.1 (Subdivision with controlled denominator)**.** *Let X be a finite n-dimensional polyhedral complex with rational structure L, integral over $\frac{1}{\mu}\mathbb{Z}$, and assume*

$$m(X, L, \mu) = k. \tag{35}$$

Then there exists a positive integer

$$v \leq \left(\left\lceil \frac{n}{2} \right\rceil !\right)^{k-1} \tag{36}$$

and a projective subdivision X' of X such that

$$L|_{X'} \text{ is integral over } \frac{1}{v\mu}\mathbb{Z}, \quad m(X', L, \mu v) = 1. \tag{37}$$

The factorial displayed above may, according to the handwritten note, be replaced by

$$\operatorname{lcm}\left\{1, \ldots, \left\lceil \frac{n}{2} \right\rceil\right\}. \tag{38}$$

Proof strategy recorded in the notes. The argument refines the subdivision process used in the theorem referred to there as "Theorem 4.1 ".

Fix a prime $p \mid k$. By the preceding lemma choose mutually compatible simplices σ_i with interior Waterman points P_i of order p whose closed stars cover the p-bad locus. Perform the mixed/star subdivision with respect to all the P_i simultaneously. The separation condition on the σ_i makes these subdivisions compatible.

By the numerical lemma, each P_i may be chosen with value bounded by

$$v(P_i) \leq \left\lceil \frac{n}{2} \right\rceil \tag{39}$$

in the estimate used in the notes. Therefore one may take the mixed denominator v to divide the least common multiple of the possible values; the notes record the coarser factorial bound as well.

It remains to verify that the maximal multiplicity decreases. If $m(\sigma) = k$, then σ lies in the star of some selected σ_i. For every simplex $\sigma' \subset \sigma$ occurring in the resulting subdivision, the Waterman-point subdivision gives

$$m(\sigma', L, \mu v) < m(\sigma, L, \mu) = k. \tag{40}$$

Thus one subdivision stage lowers the maximal multiplicity. Repeating at most $k - 1$ times gives multiplicity 1, and multiplying the denominator bounds gives the asserted estimate. □

Remark 6.2. The handwritten page 5 stresses that the principal obstacles to a universal bound are the original multiplicity k and the dimension n. For cone decompositions associated with Kummer surfaces, the notes suggest that the dimension may admit an independent bound, whereas controlling the multiplicity is expected to be harder.

7 Proposed application to Kummer surfaces

The final page formulates the intended application as follows.

Theorem 7.1 (Proposed boundedness statement)**.** *Let X be a Kummer surface associated with an abelian surface A. Assume that A admits semi-abelian reduction and that $A[2]$ is constant. Fix a multiplicity bound k.*

For the field extensions L/K for which X_L admits semi-stable reduction, define

$$d(X, k) := \min\{[L : K]\}, \tag{41}$$

where the minimum is taken over the allowed base changes satisfying the fixed multiplicity condition k.

Then there should exist a bound $B(k)$, depending only on k, such that

$$d(X, k) \leq B(k). \tag{42}$$

Remark 7.2 (Clarification of the last statement). The source page writes the theorem as a prospective bound rather than as a completed proof. The phrase "fix multiplicity k" is not defined on that page; in a finished version it should be tied explicitly to the multiplicity $m(X, L, \mu)$ of the relevant rational polyhedral/cone decomposition. Likewise, "$A[2]$ constant" should be read as saying that the finite 2-torsion group scheme is constant over the base field (subject to the characteristic hypotheses used in the surrounding project).

References

[1] G. Kempf, F. Knudsen, D. Mumford and B. Saint-Donat, *Toroidal Embeddings I*, Lecture Notes in Mathematics, vol. 339, Springer-Verlag, Berlin, 1973.